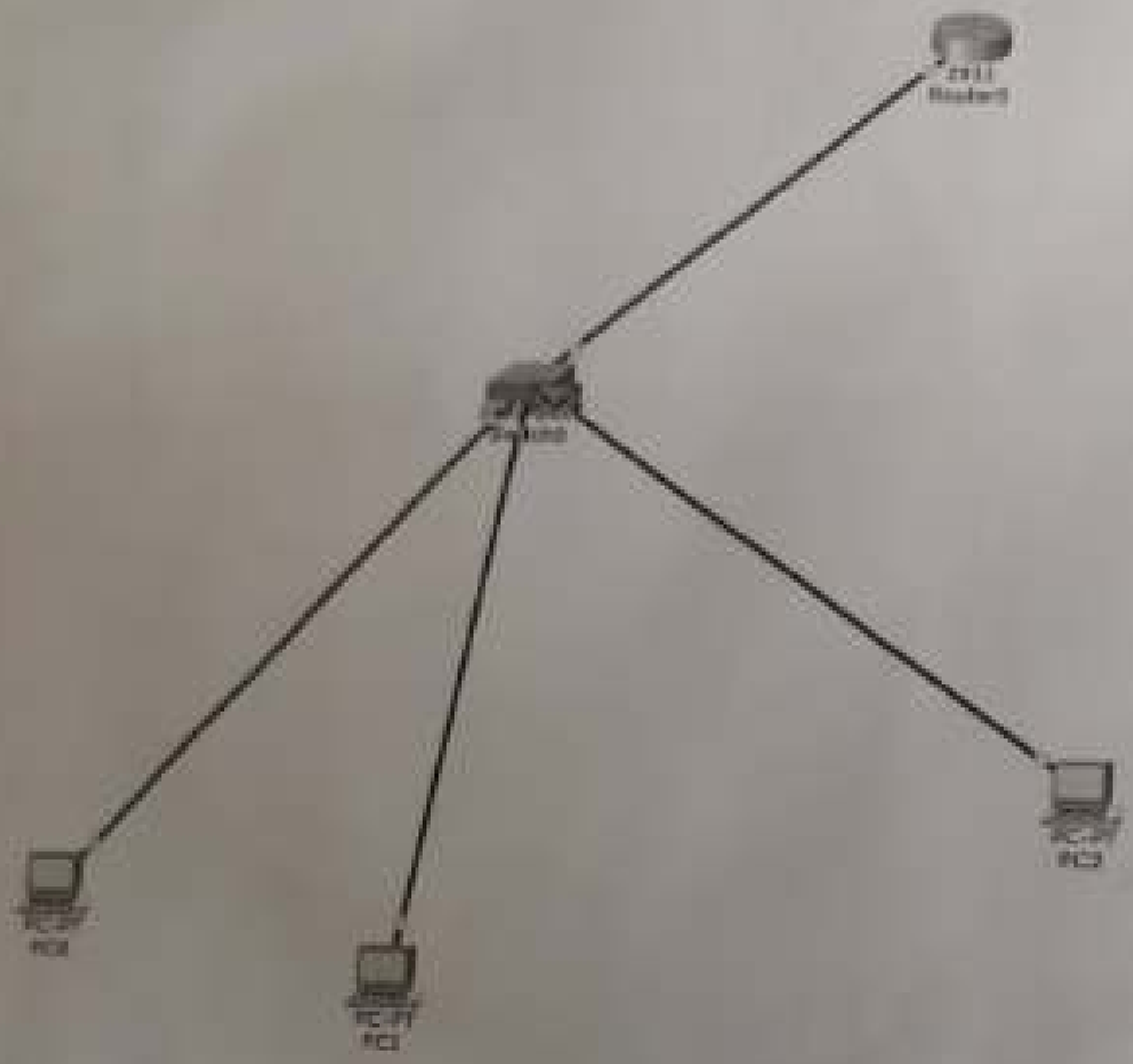


```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no

Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```

Exp No 10

Date

Routers in Cisco Packet Tracer

Aim: To configure a Router

To create a simple internetwork using Cisco Packet Tracer and observe.

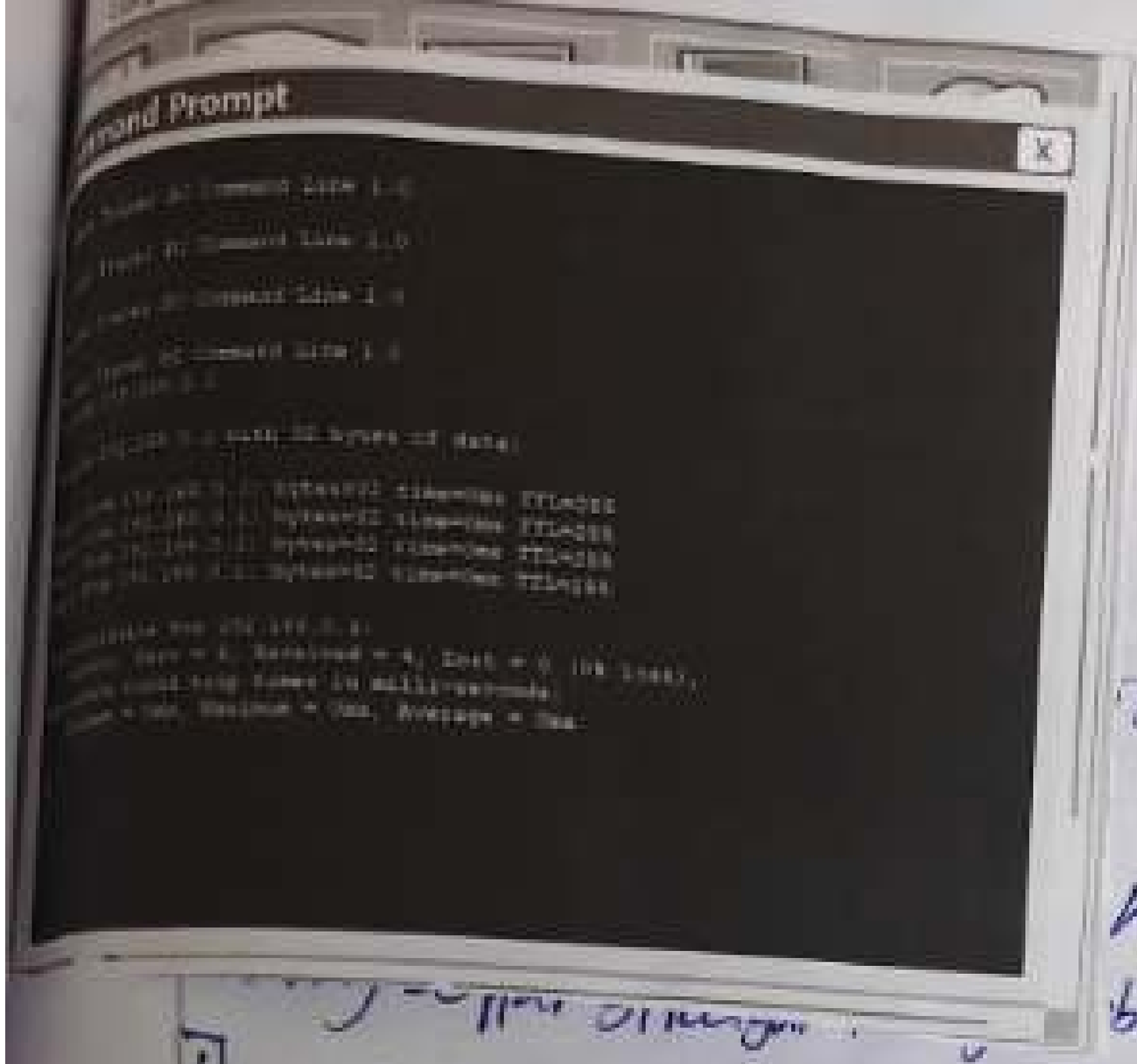
Procedure:

1. Open Cisco Packet Tracer.
2. Place one Cisco 2911 Router, two Switch and 2-4 PCs for each network
3. Connect the PCs to the switch using Copper Straight cables
4. Connect the Switches to the router using Ethernet Cables
5. Assign static IP addresses to all PCs and router interfaces
6. Configure the router interface using assigned IP addresses
7. Enable the router interface using no shutdown command
8. Verify the interface status using show ip interface brief
9. Using simple PDU to send packet from one network to
10. Select simulation mode and observe the packet flow
11. Observe that the router forwards the packet through the appropriate interface according to destination network

Result:

The Simple Internetwork was successfully design and configured using a router in CPT. Packet Transmission and routing between different networks was successfully.

```
Router#enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```




Server and

Boom Wireless Router.

WAN Port of WRT 300N

ble.

4] Connect the PC to a LAN Port of the WRT300N using a Copper Straight LAN Cable.

5] Install a wireless such WRT 300N in the laptops if required

6] Configure the WRT 300N with ZAP IP address 192.168.0.1. and subnet mask 255.255.255.0.

7] Enable the DHCP Server in wireless router

8] Connect the laptops to the received IP address automatically

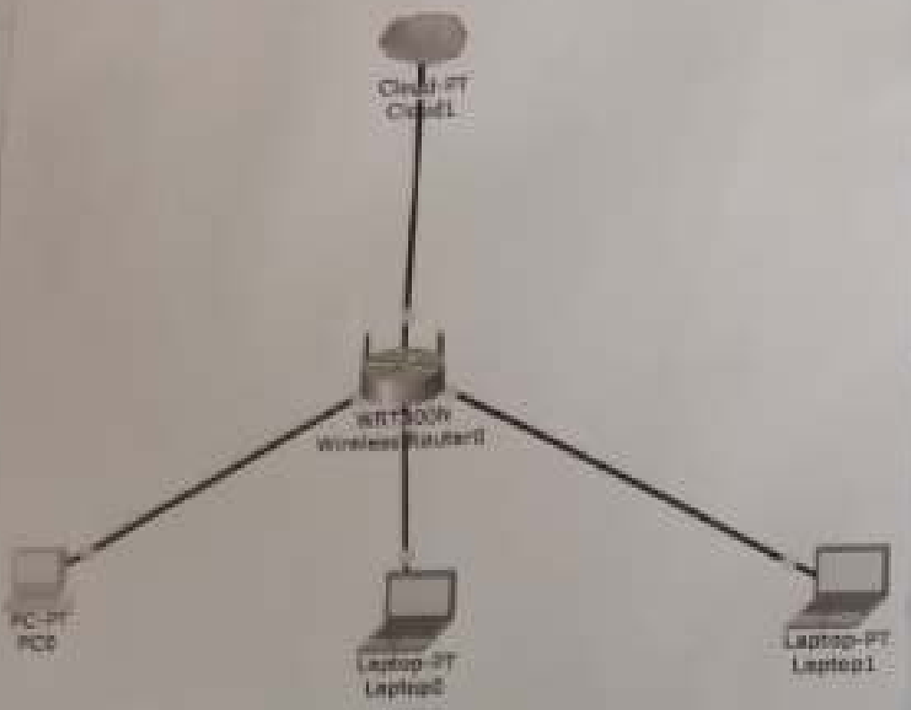
9] Verify that the devices have received IP address successfully

10] Use the Ping command to test connect between the devices

11] Use Simulation Mode to observe the packet transmission through the wireless router.

Result:-

The internetwork using a wireless router, DHCP Server and internet cloud was succesfully designed and configured. The devices obtained IP config automatically through DHCP and communication between the devices was successfully verified.




## Command Prompt

```
Packet Tracer PC Command Line 1.0
PC> ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

PC-PT
PC1

```
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.30.1 255.255.255.0

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```

5] Internetwork using Switch, Router and Ethernet cable

Aim:-

To design and configure an interface using Switch Router and Cables in Cisco Packet Tracer.

Procedure:-

1] Open Cisco Packet Tracer.

2] Place one Cisco2911 Router one switch and 3 PCs

3] Connect the Router to the Switch using a Copper Straight Through Cables.

4] Connect the Three PCs to different port of the switch using Straight Through Cables.

5] Assign an IP address to the router interface

6] Assign an IP address to all PCs

7] Configure the default gateway of each PC using R

8] Enable the router using the no shoutdown IP addr

9] Check the router interface Status using Show IP interface brief.

10] Test Connectivity between PCs using Ping Command

11] Using Simulate mode and observe the packet travel through switch and router.

Result:-

The internetwork using a router, Switch and Ethernet Cables was successfully designed and config using CPT. Communicated between the Connect devices